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A Deep Evaluation of Digital Image based Bone Cancer Prediction using Modified Machine Learning Strategy

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***Abstract*—Every single person, regardless of their age or stage in life, is at risk of developing the fatal disease known as cancer. There is a greater likelihood that cancer will affect more than one-third of the population at some point throughout their lifespan. In particular, bone cancer is a significant cause for concern in terms of health since it frequently leads to the patient's passing away. By examining the pictures obtained from X-ray, MRI, or CT scans, it is possible to identify the presence of bone cancer. The manual process requires a significant amount of effort and time-consuming expertise, and it also requires specific knowledge. As a result, the development of an automated method to differentiate between healthy bone and bone that has advanced cancer is of the utmost importance. The texture of malignant bone in the affected region is distinct from that of healthy bone in that portion of the body. It is important to note that the sample contains a number of images that exhibit morphological characteristics that are characteristic of both malignant and healthy bone. The early identification of this malignancy, including bone cancer, may be accomplished with the highest possible degree of precision through the use of medical imaging correlation in conjunction with image processing and machine learning techniques. This study proposes a strategy for identifying bone cancer by utilizing data acquired from actual clinical trials. The method is comprised of a number of phases that have the potential to increase the accuracy of disease prediction. The proposed model is designed based on Modified Machine Learning Strategy, called Elevated Learning based Bone Cancer Prediction (ELBCP), in which it evaluates the input image and predicts the bone cancer with high precision as well as the model is cross-validated with the conventional learning based classification model called Support Vector Machine (SVM) to prove the efficiency and performance of the proposed scheme.**

Keywords— Digital Image Processing, Bone Cancer, Disease Prediction, Modified Machine Learning, Elevated Learning based Bone Cancer Prediction, ELBCP, Support Vector Machine, SVM

I. INTRODUCTION

There are 206 bones in a human body. The body's muscles attach to bones, which in turn sustain the motions [1]. Fibrous tissue loaded with spongy marrow from the bone makes up bone ligaments. Tumors arising from bone cancer begin in normally functioning cells. Cancer of the bone often manifests as a tumor in the bone [2]. Over time, the tumor expands and might potentially metastasize. It has the potential to weaken bones by destroying their tissue. Multiple tests are used by the doctor to diagnose cancer. In order to identify bone cancer in humans, X-ray imaging diagnostics is employed. The rates of X-ray absorption by healthy bone and malignant bone are distinct [11].

The ability to see the shape and function of human organs through bio-medical imaging makes them a powerful diagnostic tool. Medical images are a one-of-a-kind resource for tracking the efficacy of treatment. In order to identify illness or contaminated tissue, our bodies use a range of diagnostic imaging methods. Cancer kills more people than any other disease, regardless of gender. There is hope for a complete cure for cancer if the disease can be detected early. There is a need for early detection tools for cancer nodules because cancer of the bones is brought about by the uncontrolled proliferation of bone structures. Metastatic disease occurs when the growth goes beyond the bone as well as may spread to other organs. Bone is a common site for the development of carcinomas, which are cancers that originate in epithelial cells [3].

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Various machine learning methods used to medical services based on Ultrasound, MRI, CT, histology and microscopic pictures, in which all are parts of the biomedical imaging modalities. Biomedical imaging focuses on a variety of organs, including the liver, brain, prostate, lungs, and to investigate genetic correlations. An outstanding resource for industrial scientists, medical imaging researchers, clinicians, and students is the coupling of medical imaging with machine learning [4]. Malignant cells multiply uncontrollably and metastasize (spread) to other organs in the body at a fast pace. Cancer can develop from a number of different sources, some of which can be avoided while others cannot. It is possible to avoid risk factors including smoking, high UV radiation exposure, and alcohol consumption. Genetic factors, however, cannot be changed [12]. Both hereditary and environmental variables, such as a person's lifestyle choices (such as smoking) contribute to these genetic factors. Furthermore, among the immutable variables, age stands out the most. The following figure Fig.1 shows the MRI scan of tumor-ridden bone.

Fig.1 MRI Scan of Tumor-Ridden Bone

An unusual symptom of bone cancer, which is extremely rare, is localized swelling or pain in the bone. The uncontrolled growth of aberrant cells in the bone, which can then metastasize to other organs, is the hallmark of bone cancer [5]. We anticipate a 3900% rise in new instances of bone cancer in 2022, along with 2100 fatalities. Most cases of bone cancer are seen in individuals less than 20 years old. Common modalities utilized in medical imaging include radiography, X-ray, CT scans, MRI, ultrasound, phototherapy, as well as positron emission tomography.

In addition to being easily removed surgically, the majority of benign tumors do not recur. The nuclei of malignant tumors are bigger than those of a benign tumor. Sarcomas, the medical name for bone cancer, can start in a variety of tissues throughout the body, including fibrous tissue, bones, blood vessels, and muscles. Cancers such as chondrosarcoma, osteosarcoma, pleomorphic sarcoma, ewings sarcoma, and fibro sarcoma are quite prevalent. Bone cancer is characterized by the development of tumors in the bone, which in turn hinders bone mobility and growth. The growth of an enchondroma, benign bone tumors, begins in the cartilage.

II. Related Study

Cancer strikes individuals of any age and always takes their lives. More than one-third of the population will be diagnosed with cancer at some point in their lifetimes [6]. The main goal is to pinpoint the affected region in the bone tract, that is, the aberrant growth and illness stage, by analyzing diagnostic medical imaging techniques including X-ray, CT, and PET scans. Due to the large number of slice per pixel as well as noise, the scanned visuals might not have a good resolution. To fix this, the photos must be pre-processed using a median filter to remove the noise. We will use a genetic technique to analyze specific features in the pre-processed picture, and then we will use C NN to recover them. In order to assist the doctor decides what kind of treatment to prescribe, a CNN classifier is used to classify and record the acquired images according to the patient's disease stage [6]. Results from using the proposed approach show that bone cancer is more often detected at an early stage [6].

Research in the medical field has benefited from the use of artificial intelligence [7]. Cancer detection and prognosis apps have been developed by academics in response to the accessibility of open-source healthcare information. When faced with such difficult illnesses, models based on machine learning and deep learning offer a dependable, quick, and effective solution [7]. From 2009 to 2021, papers were chosen from the Web of Sciences, EBSCO, as well as EMBASE databases using PRISMA standards. We conducted a thorough search and included all research publications that used AI-based learning methods for cancer prediction in our study. The use of both traditional machine learning as well as deep learning-based classifications for cancer prediction has resulted in 185 publications that are deemed influential. Prediction rate, accuracy, specificity, sensitivity, detection rate, dice score, precision, area undercover, recall, as well as F1-score were some of the metrics used in the survey's comparisons, which also discussed the work of other researchers and pointed out the gaps in the current literature. We looked at potential answers to five different probes. Unfortunately, cancer death rates have not decreased despite the success of several methods suggested in the literature for making predictions. Accordingly, the difficulties in cancer prognosis necessitate greater study [7].

It is possible that the oversimplification of the proportional hazard model, which assumes that the result is a linear combination of factors, makes it unsuitable for predicting the outcome of cancer patients [8]. This model is frequently employed to assess prognostic variables in cancer patients' survival rates. We implemented and validated a deep learning-based survival prediction approach in oral squamous cell carcinoma patients in this retrospective study [8]. The study included 255 patients who were eligible for analysis and had surgery in our department between 2000 and 2017. An analysis was conducted to compare DeepSurv, an algorithm for survival prediction based on deep learning, with RSF and CPH, two other popular models. With a c-index of 0.810 in the

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training set and 0.781 in the testing set, DeepSurv outperformed the other two models, RSF (0.770/0.764) and CPH (0.756/0.694). With each new feature, DeepSurv's performance got better. As a result, survival prediction powered by deep learning has the potential to enhance prediction accuracy and direct physicians towards treatment options that enhance survival while avoiding therapies that aren't essential [8].

One form of cancer, known as bone marrow cancer, grows in the bone marrow's stem cells, which are in charge of making blood [9]. Multiple myeloma (MM) and acute myeloid leukaemia (AML) are two examples of cancers that can impact the bone marrow. Traditional methods required a highly trained individual to complete the task manually, which added significant time to the process. Consequently, a more precise method of diagnosis is necessary for the provision of effective care. By enhancing a one-of-a-kind method—classification and segmentation—the proposed study has made it possible to detect more complex diseases. Despite efforts to improve cancer cell identification accuracy and reduce false alarm rates, the study's results still lean towards the bad end of the outcome spectrum, say the researchers. Adaptive multi-objective CAT algorithms, hyper parameter settings, and deep convolutional neural networks are all part of the hybrid strategy that this research proposes. You must use cell pictures that have already been processed in order to train the proposed model [9]. Using the proposed model as a basis, an Optimized Convolutional Neural Network is trained to identify the specific kind of bone tumors. After a comprehensive analysis of the SN-AM datasets, many presentation metrics were developed and evaluated, including specificity, accuracy (recall), and F1-score. Researchers found that their overall accuracy for predicting various forms of cancer was 99.45 percent. When compared to other learning models, the suggested model was far superior at distinguishing between different types of cancer cells [9].

While there is no need for a supplementary tool for bone cancer detection, early detection is the key to saving lives [10]. Computerized Tomography and Magnetic Resonance Imaging imagery show cellular heterogeneity, making it difficult for pathologists to quickly identify bone cancer. Here, several image processing methods can aid physicians in intelligently distinguishing cancerous from non-cancerous samples. When it comes to picture categorization, Deep Learning is head and shoulders above the competition among data mining and image processing methods. Popular Deep Learning systems like ResNet, AlexNet, VGG-16, and Inception-Net still have significant flaws, such as the fact that their upper-level layers mix information from lower-level layers and build results irrespective of the spatial relations and order of the picture features. On top of that, when it comes to bone cancer categorization, it shows reduced accuracy with smaller datasets. Because there is a dearth of actual bone cancer data, this study seeks to accurately categorize the disease using magnetic resonance imaging and computed tomography scans using little training data [10]. As a result, we offer a method that accomplishes the study goal by integrating with Capsule Network. When compared to previous Deep Learning approaches, the suggested framework offers superior horizontal accuracy (95.26 percentage points) in diagnosing bone cancer, according to both qualitative and quantitative analyses [10].

III. Methodology

Data preparation, feature extraction, training, as well as evaluation are the four stages that make up the study plan. One step in the procedure involves filtering and segmenting the input b one MRI picture. The next step is to utilize machine learning classifiers to forecast the category of the bone condition based on the characteristics retrieved from the segmented picture, which are statistical texture-based aspects of illnesses. The suggested extraction and classification technique's architecture is shown in the following figure Fig.2.

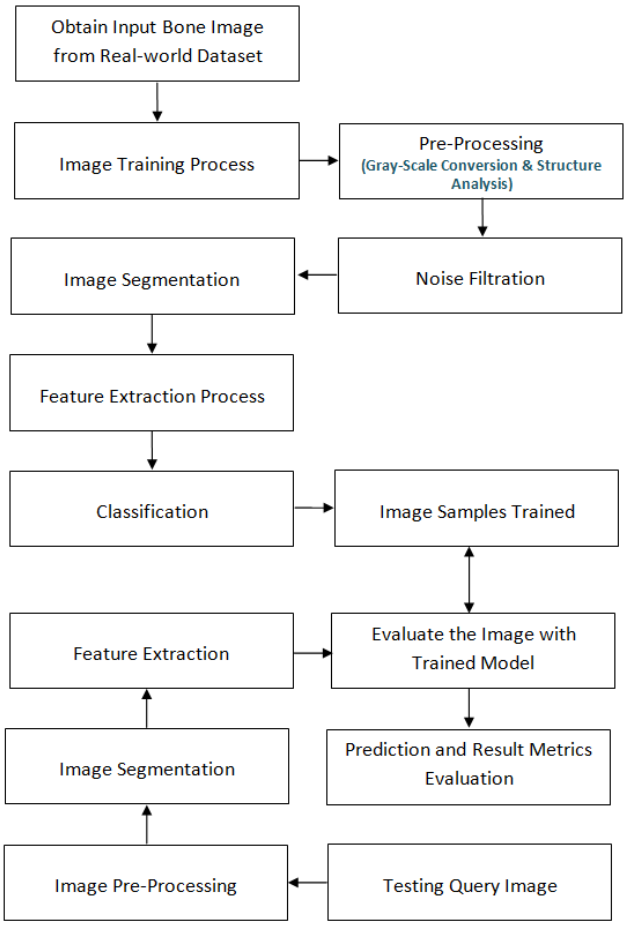

Fig.2 Proposed Architecture

Expertise is necessary for the diagnosis of bone cancer. Due to the human element involved, diagnosing bone cancer can be a time-consuming and error-prone process. It appears that the sole factor increasing the likelihood of survival for cancer patients is early identification. Using an existing machine learning technique called Support Vector Machine to detect tumors and categorize cancer, this research discusses a system termed Elevated Learning based Bone Cancer Prediction (ELBCP).

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The following figure Fig.3 shows the flow diagram of the suggested technique. An X-ray picture is sent into the system. A quick and inexpensive X-ray picture diagnosis is possible.

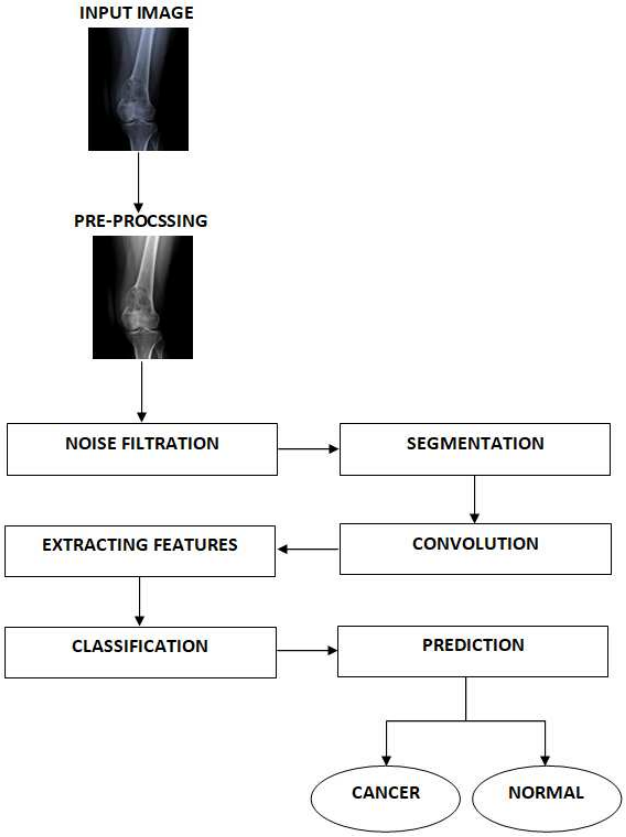

Fig.3 System Flow Diagram of Suggested Technique

(i) Preprocessing: Using a 3 × 3 median filter, the noise in the X-ray picture is eliminated. Image quality is poor. Consequently, the image gets sharpened in order to make it more intense.

(ii) Image Segmentation: After the image has undergone preprocessing, segmentation is used to identify objects in the image. The reliability of the segmentation approach is determined by the final accuracy rate. This method for identifying the object of concern is thus both reasonable and effective. Using the segmentation approach, the image is divided into pixel sets in order to extract information about the object of interest.

(iii) Feature Extraction: When describing the characteristics of a texture, a texture descriptor is quite standard. Contrast, correlation, energy, as well as homogeneity are the four texture features that we have computed from the dark level estimates of the broken image. By preserving the information in the original data, a feature extraction procedure converts the input data to numerical features.

Improvements in cancer detection and prognosis have resulted from the creation of ML models. Noise is common in medical imaging and can impact image post-processing. Consequently, it seems that the denoising of the input images is the first as well as most important step in pre-processing input images. Because losing too much noise could mask crucial features, it's critical to be precise when estimating the quantity of noise to remove. However, unintended consequences may emerge from attempting to eliminate a negligible amount of noise. Medical imaging isn't the only area that uses ML. Computer vision, data mining, and image processing all rely on feature extraction. Building a reliable bone cancer detection models that enhances the diagnostic process requires better feature extraction efficiency coupled with increased classification accuracy. To pick characteristics, this system employs mutual information statistics, a method not previously employed by bone cancer detection systems.

IV RESULTS AND DISCUSSIONS

A methodology that is adapted from machine learning is utilized by the Elevated Learning based Bone Cancer Prediction (ELBCP) method in order to tackle the classification challenge. The approach generates classification output labels by making use of data points that are adjacent to one another. ELBCP is a simplistic method that is used in the fields of statistical estimation and pattern recognition. It maintains a record of all possible cases of a class and makes an effort to classify new examples by making use of the functions of the distance, which is a measure of similarity. If k is equal to one, the case is automatically allocated to the class that received the most votes from its neighbors. If k is not equal to one, classification is easy since the class is decided by the most popular among the pixels that are closest to the case as a function of distance. Here is a list of the steps that make up the recommended algorithm:

- Determine the pairs of neighboring pixels and the index that corresponds to them, X.
- The next step is to determine the distance between the picture that was requested and the image that was used as a training sample.
- Those who are geographically closest to you have gathered and evaluated it based on the shortest distance.
- Next adjacent pixel was the source of class Y, which we gathered.
- In order to accommodate the requested number of prediction classes for the query picture, the nearest neighbor approach has been utilized for the majority of the process.

Although it has been claimed that machine learning has been successful in cancer imaging, there are still a number of obstacles and constraints that need to be solved before it can be used in c linical settings. A significant amount of data is being generated by care providers as a result of the growing demand for magnetic resonance imaging and computed tomography. Through the implementation of several standards and the Image preservation and interacting framework it has been possible to access and retrieve data in a timely manner. Data-hungry methods, such as deep neural networks, require a substantial quantity of data in order to function properly. The most significant obstacle that we have when attempting to diagnose bone

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cancer using machine learning is figuring out how to make the most effective use of the structure of medical pictures and the particular properties of medical data while we are training and constructing our model. In the field of medicine, standardized benchmarking is of utmost significance, particularly with regard to modalities and anatomic locations. Further, imaging does not serve as a standalone method for determining the severity of illness. When it comes to imaging, the molecular profile of cancer additionally serves a significant part in the diagnostic process. The potential and capability of machine learning are growing on a daily basis, but there is still some direction that has to be taken when it comes of healthcare practice. Since machine learning techniques are effective for the analysis of large amounts of complicated biological data, they may be utilized for the purpose of finding solutions to problems in the field of bioinformatics. In the field of medical image analysis, there is a need for additional progress in terms of the accuracy and predictive potential of machine learning approaches. Calculating the size, location, and level of bone cancer is possible with the use of neural networks that have undergone further development.

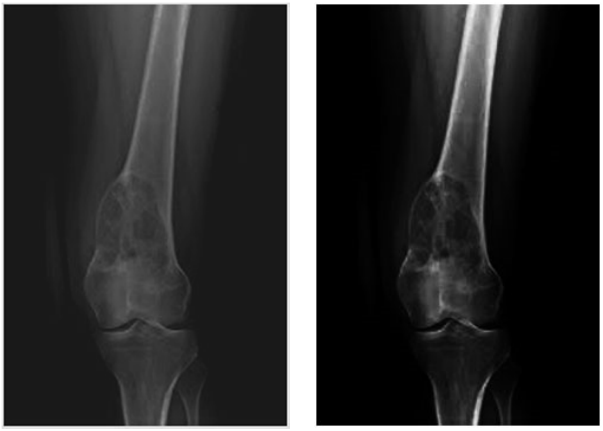
The following figure, Fig-4 and Fig-5 represents the input image and the pre-processing image stages of the proposed approach, both these functions are portrayed in conventional SVM logic also for comparison perception. The entire results of these proposed and conventional schemes are accumulated by using MATLAB image processing tool.

Fig.4 Input Image

(a) **(b)**

Fig.5 Pre-Processing (a) ELBCP and (b) SVM

The following figure, Fig-6 represents the image convolution stage results of the proposed ELBCP and conventional SVM approach.



(a) **(b)**

Fig.6 Image Convolution (a) ELBCP and (b) SVM

The following figure, Fig-7 (a) and Fig-8 (b) represents the image segmentation stages of the proposed ELBCP and conventional SVM approach.

(a) **(b)**

Fig-7 Image Segmentation (a) ELBCP and (b) SVM

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The following figure, Fig-8 represents the prediction accuracy of the proposed ELBCP and conventional SVM approach, in which the same is illustrated over the following table, Table-1 in descriptive manner.

TABLE 1 PREDICTION ACCURACY

Testing Count	SVM (%)	ELBCP (%)
5	86.34	96.15
7	86.41	96.32
9	86.11	96.14
12	86.14	96.27
15	85.32	96.41
17	85.96	96.40
18	85.53	96.45
21	85.39	96.49
23	85.24	96.54
25	85.09	96.59

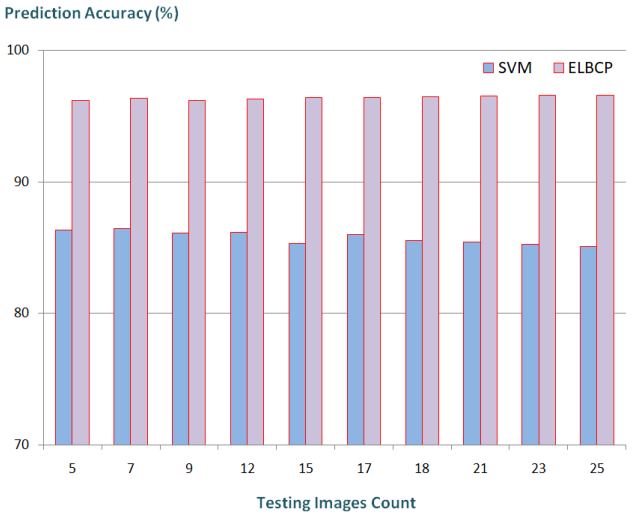

Fig. 8 Prediction Accuracy

V CONCLUSION

The proposed method integrates feature extraction with a classification model to differentiate between normal and cancerous bones. Successful detection of bone tumours has been achieved using the feature extraction and classification approach proposed here. The presence or absence of a malignancy in the input MRI can be ascertained using this method once a tumour has been identified and classified as benign or malignant. Bone cancer develops from unchecked cell growth in bone, which harms good bone tissue and can spread to other organs. The ability to recognize potential lesion tissue is crucial for pathologists to diagnose cancer. You may tweak the system's speed even further by changing other texture characteristics. In conclusion, the proposed method successfully differentiates between cancerous and healthy bone pictures. Our algorithm is less effective when presented with pictures of malignant bones compared to those of healthy bones. As a result, it may offer instantaneous second opinions to clinicians. In order to evaluate the model's performance more thoroughly, we plan to create a large dataset in the future.

REFERENCES

[1] Ashish Sharma, Dhirendra P. Yadav, et al., "Bone Cancer Detection Using Feature Extraction Based Machine Learning Model", Comput Math Methods Med, doi: 10.1155/2021/7433186, 2021.

[2] G. S et al, "Multiple Class Breast Cancer Detection Method Based on Deep Learning and MIRRCNN Model," 2 (ICICT), Nepal, 2022, pp. 981-987, doi: 10.1109/ICICT54344.2022.9850707.

[3] Abdelaal, Magdy Tobely, et al., "Bone Cancer Detection Using Particle Swarm Extreme Learning Machine Neural Networks", Journal of Medical Imaging and Health Informatics, DOI: https://doi.org/10.1166/jmihi.2019.2586, 2019.

[4] S. G et al "Automated Breast Cancer Classification based on Modified Deep learning Convolutional Neural Network following Dual Segmentation," (ICESC), Coimbatore, India, 2022, pp. 1562-1569, doi: 10.1109/ICESC54411.2022.9885299.

[5] Lianghui Zhu, Huijuan Shi, et al., "An accurate prediction of the origin for bone metastatic cancer using deep learning on digital pathological images", International Congress on Peer Review and Scientific Publication, DOI:https://doi.org/10.1016/j.ebiom.2022.104426, 2022.

[6] Sivakumar D, Manoj Krishna Hegde, et al., "BONE CANCER DETECTION USING MACHINE LEARNING", International Research Journal of Engineering and Technology, 2021.

[7] Yogesh Kumar, Surbhi Gupta, et al., "A Systematic Review of Artificial Intelligence Techniques in Cancer Prediction and Diagnosis", Arch Computat Methods Eng, https://doi.org/10.1007/s11831-021-09648-w, 2021.

[8] Dong Wook Kim, Sanghoon Lee, et al., "Deep learning-based survival prediction of oral cancer patients", Scientific Reports, https://doi.org/10.1038/s41598-019-43372-7, 2019.

[9] Manjula Devi Ramasamy, Rajesh Kumar Dhanaraj, et al., "An improved deep convolutionary neural network for bone marrow cancer detection using image processing", Informatics in Medicine Unlocked, https://doi.org/10.1016/j.imu.2023.101233, 2023.

[10] Md Abid Hassan, Md Hasibul Hasan, et al., "An Expedition to Intelligent Diagnosis of Bone Cancer and It's Direction from Capsule Network", Proceedings of the 2nd International Conference on Computing Advancements, https://doi.org/10.1145/3542954.3542981, 2022.

[11] G. Suganeshwari, R. Balakumar, et al., "DTBV: A Deep Transfer-Based Bone Cancer Diagnosis System Using VGG16 Feature Extraction", Diagnostics, doi: 10.3390/diagnostics13040757, 2023.

[12] Mengjie Fang, Zipei Wang, et al., "Predicting origin for bone metastatic cancer using deep learning-based pathology", International Congress on Peer Review and Scientific Publication, DOI:https://doi.org/10.1016/j.ebiom.2023.104449, 2023.

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